

Problem 1

Solution:

Theorem 1.22 If $a < b$ and $c < 0$ then $ac > bc$

$a < b \implies b - a \in \mathbb{R}^+$ and $c < 0 \implies 0 - c \in \mathbb{R}^+$. Now, observe that

$$\begin{aligned}(0 - c)(b - a) &= -c(b - a) \\ &= (-c)b + ac \\ &= ac - bc\end{aligned}$$

By axiom 7, $(0 - c)(b - a) \in \mathbb{R}^+$ so $ac - bc \in \mathbb{R}^+$. This in turn implies $ac > bc$

Theorem 1.23 If $a < b$, then $-a > -b$. In particular, if $a < 0$ then $-a > 0$

$a < b \implies b - a \in \mathbb{R}^+$. Now, since $b - a = -(-b) + (-a) = (-a) - (-b)$ we know $(-a) - (-b) \in \mathbb{R}^+$. This in turn implies $-a > -b$. And by direct application of this theorem, $a < 0 \implies -a > -0$. As we proved in the prior section, $-0 = 0$ so we can simplify to $-a > 0$

Theorem 1.24 If $ab > 0$ then either a, b are both positive or both negative.

We will show this by contradiction. Suppose $ab > 0$, and, without loss of generality, $a > 0$, $b < 0$. Then, since $0 < a$ and $b < 0$, by theorem 1.22, we have $0(b) = 0 > ab$. Contradiction

Theorem 1.25 If $a < c$ and $b < d$ then $a + b < c + d$

By theorem 1.18, $a < c \implies a + b < c + b$. By the same theorem $b < d \implies b + c < d + c$, or, by commutative law, $c + b < c + d$. Now, since $a + b < c + b$ and $c + b < c + d$ we can use theorem 1.17 to conclude $a + b < c + d$ as desired.

□

Problem 2

Solution:

There are 2 cases, $x = 0$ or $x \neq 0$. We will attack them directly.

If $x = 0$, then $x^2 = x \cdot x = 0 \cdot 0 = 0$. Also $x^2 + 1 = 0 + 1 = 1$. However, by theorem 1.21, $1 = (x^2 + 1) > 0$. And by theorem 1.16, we know that $x^2 + 1 \neq 0$ (because only one of $x^2 + 1 > 0$ and $x^2 + 1 = 0$ can be true).

If $x \neq 0$, then by theorem 1.20 $x^2 > 0$. Now, by theorem 1.18, $x^2 + 1 > 0 + 1 = 1$. But since we also have $1 > 0$ by theorem 1.21, we can invoke theorem 1.25 to state $x^2 + 1 > 0 + 0 = 0$. Then, by the same theorem 1.16, we know that $x^2 + 1 \neq 0$.

□

Problem 3

Solution:

Let $a < 0$ and $b < 0$. Then, by theorem 1.25, $0 = 0 + 0 > a + b$, i.e. $a + b$ is negative.

□

Problem 4

Solution:

Since $a > 0$, by theorem 1.16 $a \neq 0$. Then, by theorem 1.8, a has a reciprocal $\frac{1}{a}$ such that $a \left(\frac{1}{a}\right) = 1$. By theorem 1.21 we also know that $1 > 0$. So we have $a \left(\frac{1}{a}\right) > 0$. And now, theorem 1.24 implies that a and $\frac{1}{a}$ are both positive or both negative. Since $a > 0$, they must both be positive. As a result, $\frac{1}{a} > 0$.

For the second part, let $a < 0$. By theorem 1.16 $a \neq 0$. Then, by theorem 1.8, a has a reciprocal $\frac{1}{a}$ such that $a \left(\frac{1}{a}\right) = 1$. By theorem 1.21 we also know that $1 > 0$. So we have $a \left(\frac{1}{a}\right) > 0$. And now, theorem 1.24 implies that a and $\frac{1}{a}$ are both positive or both negative. Since $a < 0$, they must both be negative. As a result, $\frac{1}{a} < 0$.

□

Problem 5

Solution:

First off, since $a > 0$ and $b > 0$, by problem 4, $a^{-1} > 0$ and $b^{-1} > 0$. Given $a < b$ and $a^{-1} > 0$, we can invoke theorem 1.19 to state $aa^{-1} < ba^{-1}$. However, $aa^{-1} = 1 = bb^{-1}$ so we can substitute to yield $bb^{-1} < ba^{-1}$. Now, we can invoke theorem 1.19 again to state $b^{-1}bb^{-1} < b^{-1}ba^{-1}$, which simplifies to $b^{-1} < a^{-1}$. And since $b^{-1} > 0$, we can conclude $0 < b^{-1} < a^{-1}$ as desired.

□

Problem 6

Solution:

There are 4 cases. We will attack them directly.

Suppose $a < b$ and $b < c$. By theorem 1.17, $a < c$, which implies $a \leq c$.

Suppose $a = b$ and $b < c$. By substitution, $a < c$, which implies $a \leq c$.

Suppose $a < b$ and $b = c$. By substitution, $a < c$, which implies $a \leq c$.

Suppose $a = b = c$. Then, $a = c$ implies $a \leq c$.

□

Problem 7

Solution:

Since $a = c$, we have $a \leq b$ and $b \leq a$. By theorem 1.16, only one of $a = b$, $a < b$, and $a > b$ can be true. If $a < b$, then $b \leq a$ can't be true. And conversely, if $a > b$, then $a \leq b$ can't be true. Therefore, $a = b$ is the only possible outcome. And by substitution, $a = a$ (which is identically true) implies $b = c$.

□

Problem 8

Solution:

There are 3 cases. We will attack them directly.

If $a = b = 0$, then $a^2 + b^2 = (0)(0) + (0)(0) = 0 + 0 = 0 \geq 0$.

Suppose exactly one of a, b is $= 0$. Without loss of generality, say $a = 0, b \neq 0$. Then, by theorem 1.20 $b^2 > 0$. Now, since $a^2 + b^2 = (0)(0) + b^2 = 0 + b^2 = b^2$, we can state $a^2 + b^2 > 0$, which allows us to conclude that $a^2 + b^2 \geq 0$.

Suppose $a \neq 0$ and $b \neq 0$. Then, by theorem 1.20 $a^2 > 0$ and $b^2 > 0$. Finally, by theorem 1.25 $a^2 + b^2 > 0 + 0 = 0$, which allows us to conclude that $a^2 + b^2 \geq 0$.

□

Problem 9

Solution:

We will show this by contradiction. Suppose there exists a real a such that $x \leq a$ for every real x . By theorem 1.21, we know that $1 > 0$. As a result, by theorem 1.18, we know that $a + 1 > a + 0 = a$. This is a contradiction by theorem 1.16 since only one of $a + 1 > a$ or $a + 1 \leq a$ can be true.

□

Problem 10

Solution:

We will show this by simple contradiction. Suppose that there exists a positive x such that $0 < x < h$ for all positive real h . But consider what happens when $h = x$. We have a contradiction because by theorem 1.16, only one of $x < h$ and $x = h$ can be true.

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